ii. Lymphocytes

Lymphocytes are the white blood cells responsible for mediating most aspects of the immune system, with its ability to attack a wide range of infectious microorganisms. Human milk contains T and B lymphocytes.

iii. Immunoglobulins

Immunoglobulin are the difference in proteins that include all types of antibodies. Immunoglobulin are resistant to the acidity for the stomach.

iv. Lactoferrin and Vitamin B12 binding protein

Lactoferrin is an iron containing protein found both in colostrum and mature milk. It's in which the growth of staphylococcus organism and E coli by tying iron that is needed for growth. Similarly vitamin B12 binding protein present in breast milk makes vitamin B12 unavailable to pathogens that require B12 to survive in the infant's gastrointestinal track.

v. Lactobacillus bifidus factor

It is a nitrogen containing carbohydrate in human milk. It encourages the growth of microorganism, lactobacillus bifidus and produces acetic acid or lactic acid from lactose and depresses the growth of pathogenic or disease producing organism like Escherichia coli.

vi. Enzymes

Breast milk supplies enzymes like amylase, lipoprotein, lipase, bile salt, stimulated lipases, oxidases, lacto peroxidase, and leucocyte myeloperioxidase. These enzymes increase digestibility and also act as defence against microbes. Enzymes like lysozyme, peroxidase and xanthine oxidase promote cell maturation.

d) Psychological benefits

An infant derives a sense of security and belonging in the mother and child relationship from the comfort of being held than from feeding process.

Table 2.8 Calorie percentage from protein, fat and carbohydrate in human and cow's milk.

Nutrient	Breast milk	Cow's milk
Total energy (per 100 ml)	65	65
Protein	07	19
Fat	47	55
Carbohydrate	31	26

Source: Gopalan C., B.V Ramasastri and S.C Balasubramanian (2011), Nutritive value of Indian foods, NIN, ICMR, Hyderabad, India.

2.3.4 Breast feeding during illness

- Breast Feeding is to be done during illness because -
- Breast milk is the most easily digestible food for the sick baby
- Breast feeding is the best pacifier to the sick baby
- Breast milk satisfies the nutritional and fluid demands
- Breast milk offers anti-infective and immunological factors.



WORLD ALLIANCE FOR BREAST FEEDING ACTION (WABA) insists exclusive breast feeding for 6 months.

Artificial feeding

Though no milk can be real substitute for mother's milk, sometimes it is necessary to give artificial feeding. Circumstances under which artificial feeding is essential are

- The mother is suffering from serious illness, fever or infection,
- Another pregnancy intervenes during lactation

UNIT.2 PREGNANCY.indd 38 1/19/2020 2:51:34 PM